

How to write code that somebody *other than you* can read and understand

(and how to navigate code bases)

Kyle Nelli

Who am I?

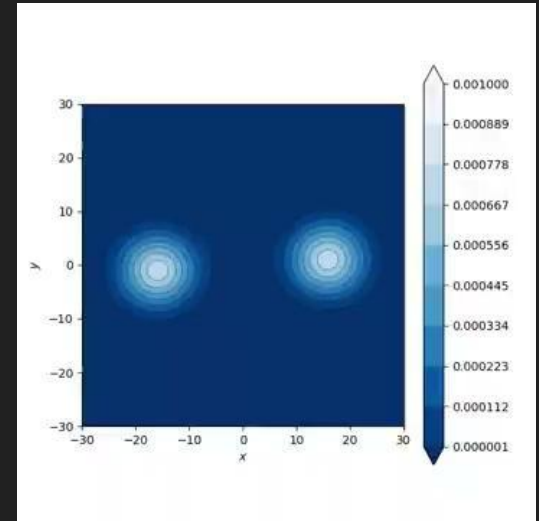
~~24001~~

No but actually...

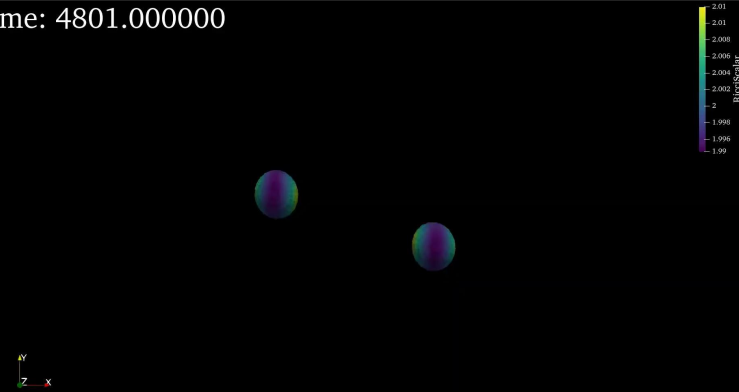


Who am I?

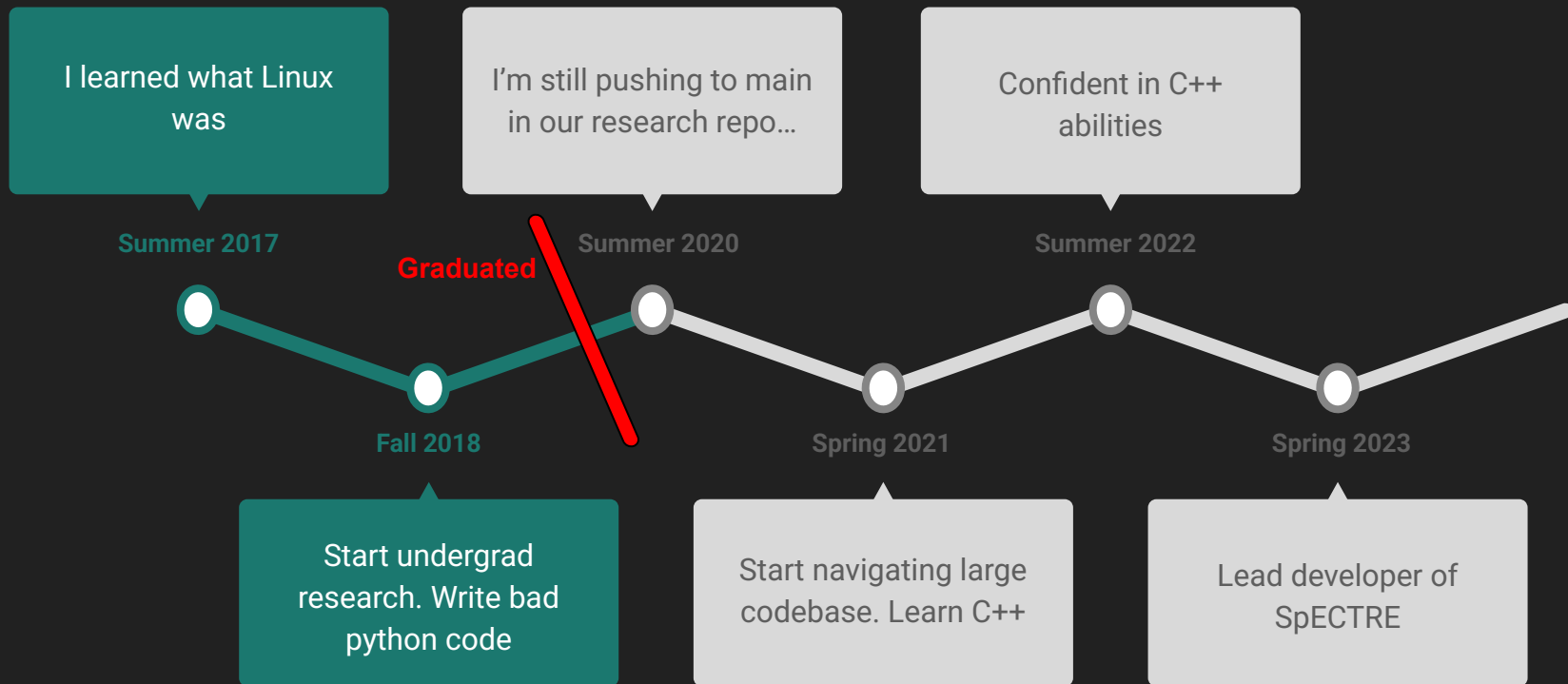
- 4th year PhD student
- From Chicago
- Undergrad at UIUC
- Cahill 360
- Simulating eXtreme Spacetimes collaboration
- SpECTRE
- NOT a software engineer...
- ...but sort of



Time: 4801.000000

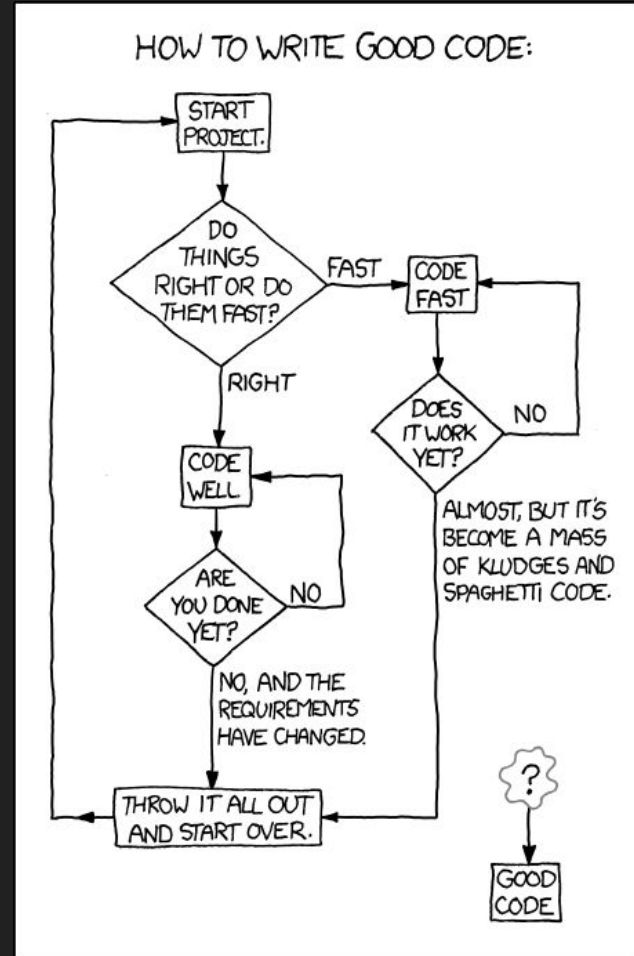


My software journey so far



Why write good code?

If it works, why does
it matter?

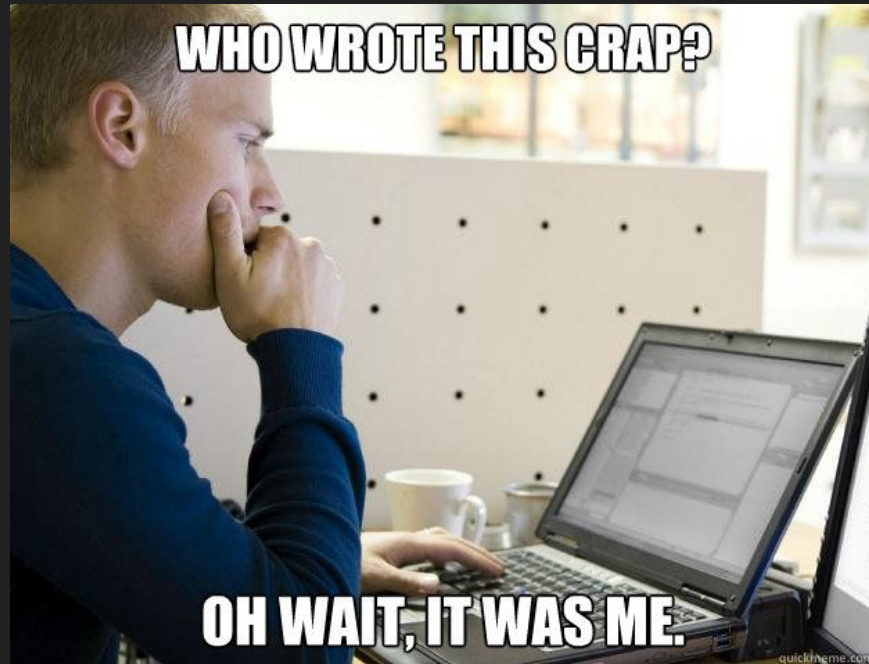


Why write good code?

People will read your code

Why write good code?

YOU will read your code



So what is good code?

1. Readable code
2. Commented code
3. Comment your code
4. Seriously comment your code
- 5. PLEASE FOR THE LOVE OF GOD, COMMENT YOUR CODE**



What is happening???????

```
CHECK(ActionTesting::number_of_queued_simple_actions<component>(  
    runner, 0) == (is_shape ? 1 : 2));  
  
if constexpr (not is_shape) {  
    ActionTesting::invoke_random_queued_simple_action<control_components>(  
        make_not_null(&runner), generator,  
        ActionTesting::array_indices_with_queued_simple_actions<  
            control_components>(make_not_null(&runner)));  
}  
  
ActionTesting::invoke_queued_simple_action<component>(  
    make_not_null(&runner), 0);  
  
CHECK(ActionTesting::number_of_queued_simple_actions<first_component>(  
    runner, 0) ==  
    (current_measurement == measurements_per_update_ ? 1 : 0));  
  
if (current_measurement == measurements_per_update_) {  
    ActionTesting::invoke_queued_simple_action<first_component>(  
        make_not_null(&runner), 0);  
}  
});
```

Ok I still
don't
understand,
but I have a
better idea
than before

```
-  
// First, there are the UpdateMessageQueue actions on the actual  
// component.  
CHECK(ActionTesting::number_of_queued_simple_actions<component>(  
    | | | runner, 0) == (is_shape ? 1 : 2));  
  
if constexpr (not is_shape) {  
    // We invoke a random measurement because during a normal simulation  
    // we don't know which measurement will reach the control system  
    // first because of charm++ communication  
    ActionTesting::invoke_random_queued_simple_action<control_components>(  
        | | | make_not_null(&runner), generator,  
        | | | ActionTesting::array_indices_with_queued_simple_actions<  
        | | | control_components>(make_not_null(&runner)));  
}  
ActionTesting::invoke_queued_simple_action<component>(  
    | | | make_not_null(&runner), 0);  
  
// If this was an update, there should be a single simple action on the  
// first component. Otherwise, no action  
CHECK(ActionTesting::number_of_queued_simple_actions<first_component>(  
    | | | | | runner, 0) ==  
    | | | | | (current_measurement == measurements_per_update_ ? 1 : 0));  
  
if (current_measurement == measurements_per_update_) {  
    ActionTesting::invoke_queued_simple_action<first_component>(  
        | | | | | make_not_null(&runner), 0);  
}  
});
```

So what are some good coding practices?

1. Descriptive variable names
2. Factor code into functions
3. Avoid excessive nesting

Examples: Descriptive variable names

```
double r = 10.0;  
double R = 1.0;  
double r1 = 5.0;
```

Ha! Sucker. You thought the big R would be the larger value? Why on earth would I do it that way? It's OBVIOUS that it's the smaller value.

```
double d = (r - r1) / (r - R);
```

Examples: Descriptive variable names

```
double outer_radius = 10.0;  
double inner_radius = 1.0;  
double test_radius = 5.0;
```

```
double fraction_of_radius =  
    (outer_radius - test_radius) /  
    (outer_radius - inner_radius);
```

Examples: Factor code into functions

```
1 magnetic_field_vector_norm = sqrt(  
2     unit_magnetic_field_vector[0] ** 2  
3     + unit_magnetic_field_vector[1] ** 2  
4     + unit_magnetic_field_vector[2] ** 2  
5 )  
6 unit_magnetic_field_vector = magnetic_field_vector / magnetic_field_vector_norm
```

```
1 normal_vector_norm = sqrt(  
2     normal_vector[0] ** 2 + normal_vector[1] ** 2 + normal_vector[2] ** 2  
3 )  
4 unit_normal_vector = normal_vector / normal_vector_norm
```

```
1 velocity_vector_norm = sqrt(  
2     velocity_vector[0] ** 2 + velocity_vector[1] ** 2 + velocity_vector[2] ** 2  
3 )  
4 unit_velocity_vector = velocity_vector / velocity_vector_norm
```

```
1 acceleration_vector_norm = sqrt(  
2     acceleration_vector[0] ** 2  
3     + acceleration_vector[1] ** 2  
4     + acceleration_vector[2] ** 2  
5 )  
6 unit_acceleration_vector = acceleration_vector / acceleration_vector_norm
```

Examples: Factor code into functions

```
1 def make_unit_vector(the_vector):  
2     norm = sqrt(the_vector[0] ** 2 + the_vector[1] ** 2 + the_vector[2] ** 2)  
3     return the_vector / norm
```

```
1 unit_magnetic_field_vector = make_unit_vector(magnetic_field_vector)
```

```
1 unit_normal_vector = make_unit_vector(normal_vector)
```

```
1 unit_velocity_vector = make_unit_vector(velocity_vector)
```

```
1 unit_acceleration_vector = make_unit_vector(acceleration_vector)
```

Examples: Avoid excessive nesting

```
if condition_1:
    print(":)")
    if condition_2:
        print(":|")
        if condition_3:
            print(":/")
            if condition_4:
                print(":(")
                if condition_5:
                    print(">:")
                    if condition_6:
                        print(">:0")
                        if condition_7:
                            print("Just...why...")
```

Other good coding practices

- Use modules/packages
- Docstrings/Documentation
- Type hints (python)
- Linting
- IDEs
- Unit-tests
- Commit often

Now it's your turn!

<https://github.com/Rhiannon-Udall/ligo-surf-software-seminars-2024>

Session_3

How to navigate a large codebase

To start off with...

You are not alone!

Tools



IDE + git



Tools



IDE +

git



= Linting + versioning + etc.

```
344 ):
345     _rich_traceback_guard = True # Hide traceback until here
346     # Determine orbital parameters
347     separation, orbital_angular_velocity, radial_expansion_velocity = (
348         initial_orbital_parameters(
```

Nils Vu, 2 months ago via PR #5933 • BBH initial data: compute initial orbital params from PN

InitialOrbitalParameters.py ~/Documents/research/spectre/bi

from typing import Optional, Sequence

6

7 import numpy as np

8 from scipy.optimize import minimize

9

10 logger = logging.getLogger(__name__)

11

12

13 def initial_orbital_parameters(

14 mass_ratio: float,

15 dimensionless_spin_a: Sequence[float],

16 dimensionless_spin_b: Sequence[float],

17 eccentricity: Optional[float] = None,

18 mean_anomaly_fraction: Optional[float] = None,

19 separation: Optional[float] = None,

20 orbital_angular_velocity: Optional[float] = None,

21 radial_expansion_velocity: Optional[float] = None,

22

23 mass_ratio=mass_ratio,

24 dimensionless_spin_a=dimensionless_spin_a,

25 dimensionless_spin_b=dimensionless_spin_b,

26 separation=separation,

27 orbital_angular_velocity=orbital_angular_velocity,

28 radial_expansion_velocity=radial_expansion_velocity,

29 eccentricity=eccentricity,

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Where to look

- [Documentation/wiki](#)

Where to look: Documentation

SpECTRE v2024.06.18

Q Search

Introduction

Installation

User Tutorials

A Hitchhiker's Guide to Running SpECTRE

Running and Visualizing

OrientationMap

Domain Creation

Using SpECTRE's Python modules

Command-line interface (CLI)

Running CCE

Setting up checkpoints and restarts

Events and triggers

IMEX

User Tutorials

- [A Hitchhiker's Guide to Running SpECTRE](#) - A brief intro to getting started with SpECTRE, building and rebuilding an executable with new features, and visualizing the output
- [Running and Visualizing](#) - How to run an existing executable and visualize its output
- [OrientationMap](#) - How to determine orientations
- [Domain Creation](#) - How to set up the initial domain
- [Using SpECTRE's Python modules](#) - How to install and work with the Python bindings and other Python modules provided by SpECTRE.
- [Command-line interface \(CLI\)](#) - How to use the command-line interface
- [Running CCE](#) - How to get started running stand-alone CCE with externally produced data
- [Setting up checkpoints and restarts](#) - How to set up a SpECTRE executable for checkpointing and restarting
- [Events and triggers](#) - How to specify code to run in an input file
- [IMEX](#) - How to add IMEX support to an evolution system

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Where to look: Documentation

control_system::UpdateControlSystem< ControlSystem > Struct Template Reference

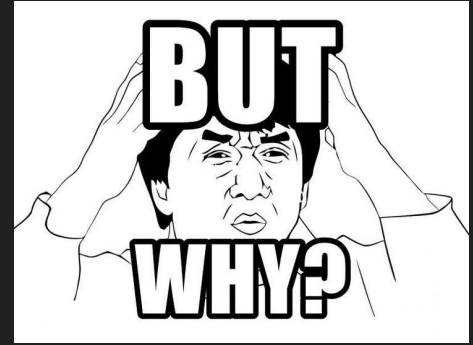
Functor for updating control systems when they are ready. [More...](#)

```
#include <UpdateControlSystem.hpp>
```

The algorithm to determine whether or not to update the functions of time is as follows:

1. Ensure this control system is active. If it isn't, end here and don't process the measurements.
2. Increment the current measurement stored by `control_system::Tags::CurrentNumberOfMeasurements`. This keeps track of which measurement we are on out of `control_system::Tags::MeasurementsPerUpdate`.
3. Calculate the control error and update the averager (store the current measurement). If the averager doesn't have enough information to determine the derivatives of the control error, exit now (this usually only happens for the first couple measurements of a simulation). If this is `size` control, an extra step happens after we calculate the control error, but before we update the averager. See `control_system::size::update_averager` for this step.
4. If the `control_system::Tags::WriteDataToDisk` tag is set to `true`, write the time, function of time values, and the control error and its derivative to disk.
5. Determine if we need to update. We only want to update when the current measurement is equal to the number of measurements per update. If it's not time to update, exit now. Once we determine that it is time to update, set the current measurement back to 0.
6. Compute the control signal using the control error and its derivatives and update the damping timescales in the [TimescaleTuner](#). If this is `size` control, there is an extra step after we update the damping timescale. See `control_system::size::update_tuner` for this step.
7. Calculate the new measurement timescale based off the updated damping timescales and the number of measurements per update.
8. Determine the new expiration times for the `domain::Tags::ExpirationOfTime` and

Where to look: Documentation?????



amr::Info< VolumeDim > Struct Template Reference

Information about an element that is communicated by AMR actions.

```
#include <Info.hpp>
```

Where to look

- Documentation/wiki
- Tests

Where to look: Tests

```
import h5py
import numpy as np
import numpy.testing as npt

import spectre.IO.H5 as spectre_h5
```

```
# Test whether a dat file can be added correctly
def test_insert_dat(self):
    with spectre_h5.H5File(file_name=self.file_name, mode="a") as h5file:
        datfile = h5file.insert_dat(
            path="/element_data", legend=["Time", "Value"], version=0
        )
        self.assertEqual(datfile.get_version(), 0)
```

```
# Test whether a cce file can be added correctly
def test_insert_cce(self):
    with spectre_h5.H5File(file_name=self.file_name, mode="a") as h5file:
        ccefile = h5file.insert_cce(path="/cce_data", l_max=2, version=0)
        self.assertEqual(ccefile.get_version(), 0)
```

```
# Test whether data can be added to the dat file correctly
def test_append_dat(self):
    with spectre_h5.H5File(file_name=self.file_name, mode="a") as h5file:
        datfile = h5file.insert_dat(
            path="/element_data", legend=["Time", "Value"], version=0
        )
        datfile.append(self.data_1)
        outdata_array = np.asarray(datfile.get_data())
        npt.assert_array_equal(outdata_array[0], self.data_1_array)
```

Where to look

- Documentation/wiki
- Tests
- Error messages

Where to look: Error messages

The screenshot shows the Visual Studio Code interface with the file `InitialData.py` open. The editor displays the following code:

```
155
156 # Resolve directories
157 if pipeline_dir:
158     pipeline_dir = Path(pipeline_dir).resolve()
159 assert segments_dir is None, (
160     "Initial data generation doesn't use segments at the moment. Specify"
161     " '--run-dir' / '-o' or '--pipeline-dir' / '-d' instead."
162 )
163
164 if evolve:
165     assert pipeline_dir is not None, (
166         "Specify a '--pipeline-dir' / '-d' to evolve the initial data."
167         " Don't specify a '--run-dir' / '-o' because it will be created in"
168         " the 'pipeline_dir' automatically."
169     )
170     assert run_dir is None, (
171         "Specify the '--pipeline-dir' / '-d' rather than '--run-dir' / '-o'"
172         " when evolving the initial data. Directories for the initial data,"
173         " evolution, etc will be created in the 'pipeline_dir'"
174         " automatically."
175     )
```

The terminal shows the command execution and the resulting error:

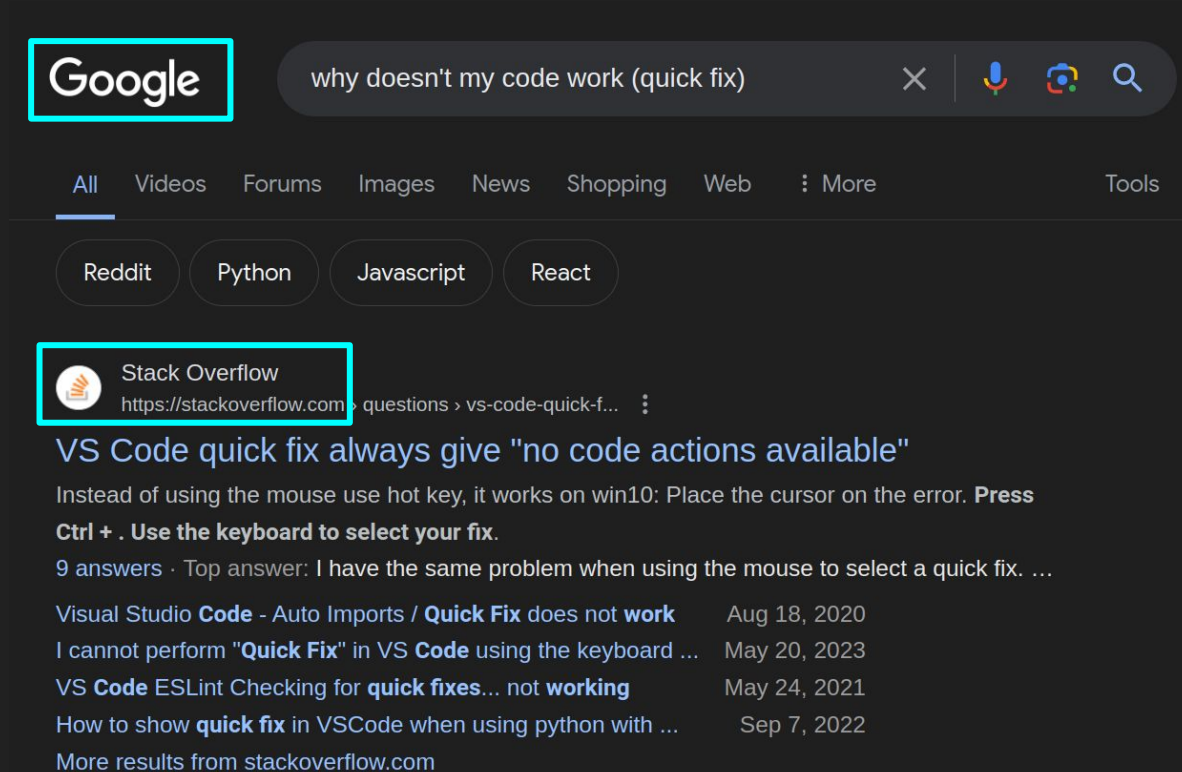
```
knelli@jarrus:~/Documents/research/spectre$ spectre bbh generate-id -q 1 --chi-A 0 0 0 --chi-B 0 0 0 -D 4 -w 0 -a 0 --evolve
[11:26:45] WARNING The BBH pipeline is still experimental. Please review the generated input files.
Traceback (most recent call last)
  /home/knelli/Documents/research/spectre/build/bin/python/spectre/Pipelines/Bbh/InitialData.py:164 in generate_id
> 164 | | assert pipeline_dir is not None, (
AssertionError: Specify a '--pipeline-dir' / '-d' to evolve the initial data. Don't specify a '--run-dir' / '-o' because it will be created in the 'pipeline_dir' automatically.
knelli@jarrus:~/Documents/research/spectre$
```

The error message in the terminal is highlighted in yellow, and the corresponding code in the editor is highlighted in red. The search bar in the top left corner shows the search query "Specify a '--pipeline-dir' / '-d' to evolve" and the result "1 result in 1 file - Open in editor".

Where to look

- Documentation/wiki
- Tests
- Error messages
- The place with all the answers...

Where to look: The places with all the answers



The image shows a Google search interface. The search bar contains the text "why doesn't my code work (quick fix)". Below the search bar, there are tabs for "All", "Videos", "Forums", "Images", "News", "Shopping", "Web", and "More". There are also buttons for "Reddit", "Python", "Javascript", and "React". The search results show a link to a Stack Overflow question titled "VS Code quick fix always give 'no code actions available'". The question text says: "Instead of using the mouse use hot key, it works on win10: Place the cursor on the error. **Press Ctrl + . Use the keyboard to select your fix.**". There are 9 answers, and the top answer is "I have the same problem when using the mouse to select a quick fix. ...". Below the answers, there are several related questions listed with their dates.

Google

why doesn't my code work (quick fix)

All Videos Forums Images News Shopping Web : More Tools

Reddit Python Javascript React

Stack Overflow
<https://stackoverflow.com/questions/vs-code-quick-f...>

VS Code quick fix always give "no code actions available"

Instead of using the mouse use hot key, it works on win10: Place the cursor on the error. **Press Ctrl + . Use the keyboard to select your fix.**

9 answers · Top answer: I have the same problem when using the mouse to select a quick fix. ...

Visual Studio **Code** - Auto Imports / **Quick Fix** does not **work** Aug 18, 2020

I cannot perform "**Quick Fix**" in VS **Code** using the keyboard ... May 20, 2023

VS **Code** ESLint Checking for **quick fixes**... not **working** May 24, 2021

How to show **quick fix** in VSCode when using python with ... Sep 7, 2022

More results from stackoverflow.com

I'm stuck...what do I do?

- Always ask for help
- How long before I should ask for help?
 - Varies, but couple hours at most
 - Eat something
 - Maybe sleep on it
- How should I ask for help?
 - Be specific
 - Bad
 - “My run failed”
 - “I got an error”
 - Good
 - “I was trying to generate initial data on the cluster with these parameters [insert params] but I got an error saying [paste error]”

Happy coding! <3